

NAM KHANH (KEI) TRAN

(+61) 425 439 747 | Sydney, NSW | namkhanh.tran2310@gmail.com

[linkedin.com/in/kei-tran](https://www.linkedin.com/in/kei-tran) | github.com/Lolmantran | lolmantran.github.io/Portfolio/

SUMMARY

Full-Stack Software Engineer with 1+ years of experience delivering reliable, efficient systems, balancing new feature development with maintaining and improving production applications. Experienced in deploying cloud-based systems on AWS (AWS Certified Solutions Architect Associate). Focused on clean architecture and long-term maintainability, with an interest in AI-augmented systems.

EDUCATION

University of New South Wales	Sep 2024 – Aug 2027 (Expected)
Bachelor of Computer Science, Major in Artificial Intelligence (Dean’s Honours List 2025)	WAM: 86/100
- Remarkable High Distinction course: Computer Systems, Web Frontend Programming, Introduction to Artificial Intelligence, Software Engineering Fundamentals, Programming Fundamentals - Distinction course: Data Structure and Algorithm, Object Oriented Programming	

WORK EXPERIENCE

UNSW Taste of Research Software Engineer (Research Scholarship)	Sydney, NSW Jun 2026 – Present
- Engineered a modular image-transmission pipeline (JPEG → bit-stream packing → BPSK modulation → AWGN channel → decode) with configurable quality and noise parameters for reproducible runs. - Built an automated benchmarking harness comparing four reconstruction methods (JPEG baseline, SRCNN, Cycle-CNN, Deep JSCC) across a full noise sweep, auto-generating PSNR comparison charts and metrics. Refactored an ad-hoc codebase into reusable, well-structured modules and a self-contained pipeline, cutting a full experiment re-run from many manual steps to a single command.	
FPT Software Software Engineering Intern (Full-time)	Hanoi, Vietnam Nov 2025 – Apr 2026
FPT Software – Global Healthcare Facility, a department specialising in developing software solutions for healthcare providers.	
Designed and implemented an end-to-end AI document pipeline (AWS Textract/OCR → chunking → vector embeddings), and built a RAG-based retrieval system enabling LLMs to generate source-grounded answers with citations, improving AI agent accuracy by 50–70%. Built a Queue Management System within a 5-year-old microservice architecture, implementing automated check-in flows, queue state transitions, and vital signs capture, reducing patient check-in time and accelerating overall processing by 80%. Integrated AI agents into clinical workflows using NextJS and ExpressJS, enabling symptom-based decision support, clinical note extraction from unstructured data, and automated prescription generation, improving clinician efficiency. Designed and implemented real-time patient queue tracking using Node.js WebSockets, enabling live drag-and-drop patient management across service rooms. Designed a scalable PostgreSQL database schema for multi-location hospital systems, supporting hierarchical structures, and successfully deployed in 2 hospitals in Malaysia. Practised TDD, achieving 95% code coverage using Jest unit tests and Playwright E2E tests.	
Apus Web Software Engineering Intern (Part-time)	Hanoi, Vietnam May 2024 – Dec 2024
Apus Web is a web development outsourcing company running a structured program for junior developers to contribute to real client projects.	
Built a full-stack e-commerce application in React.js, meeting company quality standards, covering product listing, cart, and checkout flows Developed RESTful APIs with Node.js and MongoDB, handling data modelling and CRUD operations for the full product Enhanced authentication security on a live client project by implementing JWT-based login, receiving and incorporating direct client feedback - Gained hands-on experience with frontend fundamentals, including HTML, CSS, and Bootstrap in a production outsourcing environment	

PROJECT

VietLearn (NextJs, NestJs, Prisma, Docker, Railway)
Demo: https://vietlearn-frontend.vercel.app/
Developed a full-stack AI Vietnamese learning platform with a Next.js 16 (App Router) frontend and NestJS backend — featuring spaced-repetition flashcards, multiple-choice quizzes, OpenAI-powered AI tutor chat, real-time Recharts dashboard with XP/streak gamification, and Google OAuth authentication via NextAuth v5 with server-side protected routing middleware Architected and deployed a production-grade system end-to-end using PostgreSQL (Supabase) with Prisma ORM, TanStack React Query v5 for server state, JWT-based auth with cookie syncing, and a full CI/CD pipeline — GitHub Actions for automated lint/build checks, frontend hosted on Vercel, and a Dockerized NestJS backend deployed on Railway.

SKILLS

AI & Automation: Claude Code, GitHub Copilot, MCP Server, Agent Orchestration, Prompt Engineering, OpenCV, TensorFlow, Numpy Languages: JavaScript (ES6+), TypeScript, Python, Java, C/C++, Assembly Databases: PostgreSQL, MongoDB, Redis. Frontend & Framework: React, Next.js, Node.js, Redux, Material-UI, Tailwind, Bootstrap Backend & Infrastructure: NodeJS (NestJS, ExpressJS), AWS (EC2, S3, Lambda), Docker, Java Spring Testing & Quality Assurance: Jest, Playwright, TDD
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AWARDS AND CERTIFICATIONS

UNSW Taste of Research Scholarship AWS Certified Solution Architecture Associate (SAA-C03) - Samsung Innovation Campus, Coding & Programming with C++ course - DevSoc Flagship Hackathon, Honourable mention	Jun, 2026 Apr, 2026 May, 2024 Oct, 2025
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